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De-Orbit Maneuver Demonstration Results of Micro-Satellite ALE-1 with a Separable Drag Sail

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Abstract: ALE-1, a micro-satellite created for the demonstration of artificial shooting stars, required orbital descent before mission execution due to safety aspects in orbit. ALE-1 utilized a drag sail called SDOM (Separable De-Orbit Mechanism) for a passive de-orbit maneuver, which was successfully completed, lowering the orbit from about 500 km down to about 400 km. This paper summarizes the detailed history of satellite operation and the results of the de-orbit maneuver demonstration during the past three years. Although the SDOM sail faced difficulty in keeping the desired deployed shape of the drag sail due to mechanical troubles, by letting the sail be a drag flag instead, it could still deliver a meaningful de-orbit performance to allow the satellite to successfully lower the orbit as planned. The de-orbit effect of the drag flag was evaluated using comparisons between orbit propagation simulations and the actual orbit transition flight data provided in the form of TLE (Two-Line Element) sets. Through this study, it is demonstrated that the SDOM can provide orbit transfer capabilities for satellites. Furthermore, the de-orbit performance of the drag flag can be evaluated, which could be an important reference for the future implementation of de-orbit devices to solve space debris problems.

Keywords: micro-satellite; drag sail; de-orbit; time-of-flight camera

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1. Introduction

ALE-1 is a micro-satellite jointly developed by Tohoku University and ALE Co., Ltd. (Tokyo, Japan), and it was launched by Japanese Epsilon Launch Vehicle No. 4 on 18 January 2019 as part of the first implementation of JAXA's Innovative Satellite Technology Demonstration Program [1,2]. The mission of ALE-1 is to demonstrate the technology of artificial shooting star generation, and it is still operational in orbit. ALE-1 required a descent of about 100 km from the initial orbit altitude of about 500 km due to safety aspects in orbit, which was successfully accomplished with a passive drag sail de-orbit device called Separable De-Orbit Mechanism (SDOM). SDOM is a passive de-orbit system that uses atmospheric drag in the low Earth orbit, and it was jointly developed by Nakashimada Engineering Works Ltd. and Tohoku University. Although it takes a long time to de-orbit, it is a simple thin-film deployment system that is actuated by mechanical strain energy, and it is expected to be a solution to the space debris problem, which has been a significant concern in recent years [3,4].

We have reported on the development and operational progress of SDOM and ALE-1 in various ways. In particular, in 2022, we published a paper on the status of system trouble that occurred in SDOM and its impact on the orbit descent [5]. Two of the four corners of the square-shaped drag sail of SDOM were unexpectedly loosened, resulting in a drag flag. This system trouble seemed to decrease SDOM's performance as a de-orbit mechanism, and we discussed the prediction of a possible significant delay in the orbit descent.

Despite the trouble, however, SDOM was subsequently able to fulfill its role in orbit descent and revealed that the effect of the trouble was minimal. Moreover, SDOM was successfully separated from the ALE-1 satellite at the end of 2022 as intended, allowing the micro-satellite to stay in the target orbit for a longer time period. In this paper, we review the history and results of the orbital operation of ALE-1 in terms of various aspects and evaluate the descent performance of the SDOM based on the comparisons between orbit propagation simulations and the actual orbit transition record provided in the form of TLE (Two-Line Element) history.

2. Mission and System

2.1. ALE-1

Figure 1 shows the appearance of the flight model of the micro-satellite ALE-1 together with the positions of the related components, and Table 1 provides its specifications. The main mission of ALE-1 is to demonstrate the technology of an "artificial shooting star" generation by ejecting small metal pellets from orbit to Earth, where interaction with the atmosphere will cause a luminous phenomenon. This technological demonstration has two aspects: entertainment and scientific observation. The entertainment aspect involves the development of a service that allows people to enjoy shooting stars from any ground position at any given time. The scientific observation aspect aims to gather data on luminous phenomena in the upper atmosphere, which can be used to investigate the characteristics of the upper atmosphere and the re-entry behavior of small objects [6–9].

Table 1. Specifications of ALE-1.

Parameters		Values
Satellite	Mass [kg]	68.16
	Dimensions [mm]	440 × 500 × 539
SDOM	Mass [kg]	3.88
	Dimensions [mm] (Stored configuration)	277 × 211 × 222
	Film size [mm]	2500 × 2500
	Thickness of the film [μm]	25

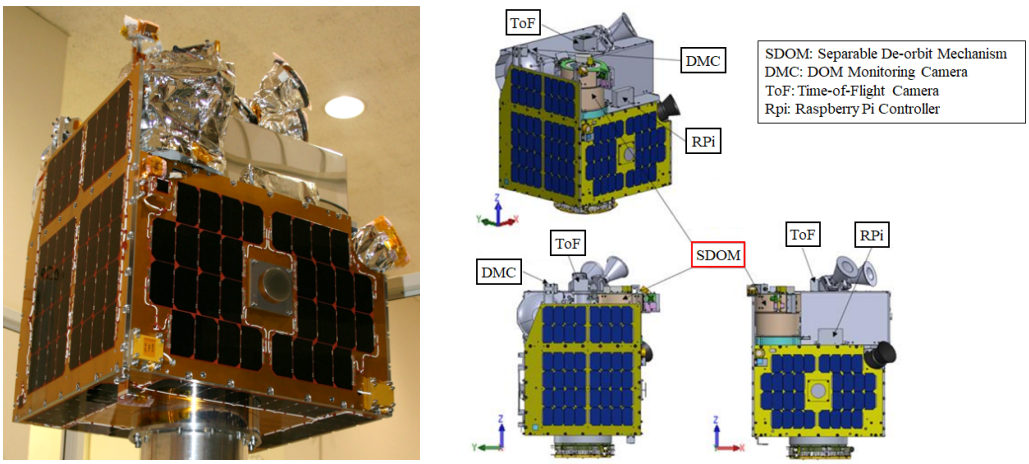


Figure 1. Micro-satellite ALE-1 and its component configuration.

In order to prevent collisions with other spacecraft in orbit, it is crucial to ensure that the pellets are carefully ejected. This is especially important for the International Space Station (ISS), which operates at an altitude of 400 km and houses critical experimental facilities and astronauts. To minimize the risk of any pellet impact with the ISS, ALE-1 needed to transfer from its initial orbit at an altitude of 500 km to a lower altitude below 400 km prior to commencing its main mission [10–12].

As the orbit transfer device, we installed an SDOM, which will be described in detail in the next section. The SDOM utilized a sail to increase the atmospheric drag to lower the orbit altitude and separated the sail section after ALE-1 had descended about 100 km, allowing it to remain in the lowered orbit for an extended period.

2.2. SDOM: Separable De-Orbit Mechanism

SDOM is a passive de-orbit system that was specifically developed for ALE-1. This system uses an aluminized polyimide thin film as an atmospheric drag sail and represents a new model of the de-orbit mechanism (DOM) that Nakashimada Engineering Works, Ltd. and Tohoku University have been collaboratively developing since 2010 [13,14]. The appearance of the deployed DOM is illustrated in Figure 2. The SDOM system drags down satellites using the atmospheric drag that acts on the thin film, allowing satellites to re-enter the Earth's atmosphere. Several models of DOM have been already operated in orbit, providing us with important engineering findings [15,16].

SDOM consists of the following components:

- A cylindrical container to ensure safety during rocket launch;
- A thin film deployment mechanism that acts as a drag sail to lower the orbit;
- A boom to keep the thin film away from the satellite so that it does not block the antenna or solar panels;
- A system to separate the drag sail from the container after reaching a predetermined target altitude.

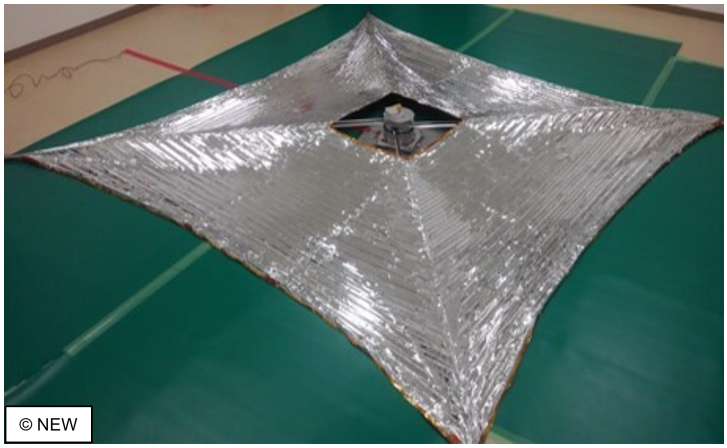


Figure 2. Deployed configuration of DOM: De-Orbit Mechanism.

SDOM is designed to have five operational phases to ensure its proper functioning, as depicted in Figure 3 [5]. Phase 0 represents the safe storage of all components within the cylindrical body without deployment. In phase 1, the lid is opened. In phase 2, the boom is extended to position the drag sail mechanism 2.5 m away from the satellite's main structure. In phase 3, a 2.5 m × 2.5 m thin film is deployed to initiate de-orbit. Upon reaching a predetermined target altitude (in the case of ALE-1, less than 400 km), the film and boom are separated on command from the ground station to prolong the satellite's orbital lifetime at that altitude (phase 4). The cylindrical container remains attached to the satellite body.

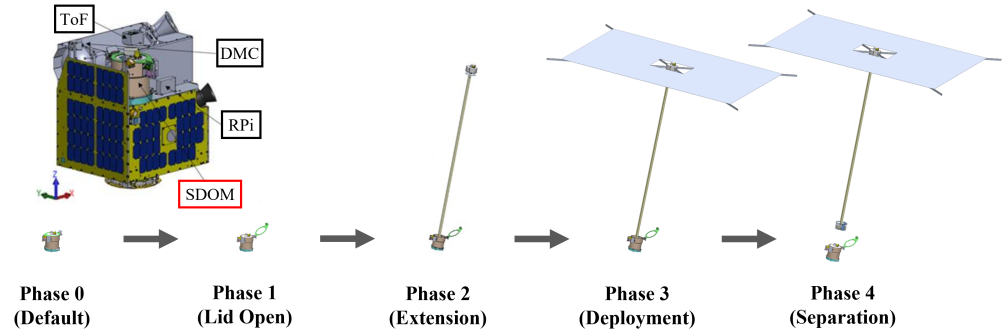


Figure 3. Operational phases of SDOM.

3. History of Operation

3.1. Mission History

In this section, the operational history of ALE-1 and SDOM is discussed using images from the TOF (Time-of-Flight) camera onboard ALE-1. The mission history is shown in Figure 4, and the corresponding TLE history of ALE-1 is shown in Figure 5.

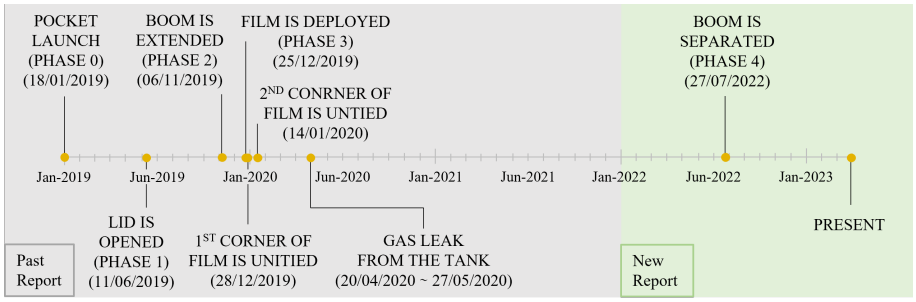


Figure 4. Mission history of ALE-1 and SDOM.

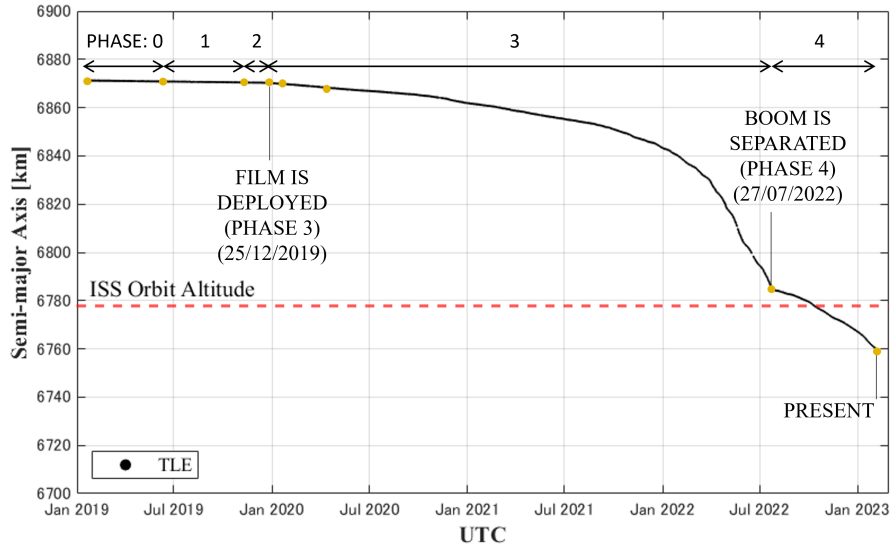


Figure 5. Orbital altitude history of ALE-1 based on TLE: Two-Line Element. The semi-major axis of the orbit is plotted together with the mission history.

The separated SDOM sail is unlikely to generate additional space debris, as its area-to-mass ratio is large enough to de-orbit itself for a very short orbital period, ultimately being burned up during the re-entry. One reference for this technology is the 1U-sized CubeSat FREEDOM, which was operated in 2017 by Nakashimada Engineering Works, Ltd. and Tohoku University [17]. FREEDOM carried a DOM with a film size of 1.5 m × 1.5 m. Based

on the orbit history of FREEDOM, it took about 22 days to re-enter the Earth's atmosphere from an altitude of 400 km [13]. According to this background, the separated part of the SDOM is expected to re-enter the atmosphere within a few days.

A detailed description of the mechanisms and satellite subsystems of ALE-1 can be found in [7,18,19]. Each phase transition is initiated by stored commands that are uplinked from the ground station after the required status checks for each phase transition have been verified [11].

After ALE-1 entered orbit on 18 January 2019, the satellite was operated and underwent orbital functional verification. The SDOM mission started on 11 June 2019 when its lid was opened. Shortly thereafter, the boom extension was initiated, but it could not be confirmed as intended. After 5 months of regular observations, the boom extension was finally confirmed on 6 November 2019, which initiated phase 2. The deployment of the thin film was postponed for an additional 50 days to allow for the detailed evaluation of the gravity gradient effects acting on the satellite system. The film was then deployed on 25 December 2019 [5].

Based on the initial estimation, ALE-1 with the SDOM deployed was expected to descend to the ISS orbital altitude in about 650 days [18]; in contrast with the estimation, however, the deployed SDOM experienced a series of problems. On 28 December 2019, one of the film's four corner connections was found to be damaged, which left only half the area of the film effective for drag. Approximately two weeks later, another corner connection was lost, effectively changing the film from a drag sail to a drag flag. This issue is discussed in more detail in the next section. In addition, between 20 April 2020 and 27 May 2020, a gas leak occurred from a tank installed for the artificial shooting star mission; the gas leak was caused by a malfunction of the gas output control system, and the malfunction was resolved when the control system was restarted. After that, operations went smoothly, and on 27 July 2022, when the orbital altitude was confirmed to be below 400 km, the sail section of SDOM was separated from ALE-1. The SDOM separation was observed by the DMC (DOM Monitoring Camera) as illustrated in Figure 6. ALE-1 completed its descent, albeit about a year later than predicted, and is still flying today, maintaining its orbital altitude.

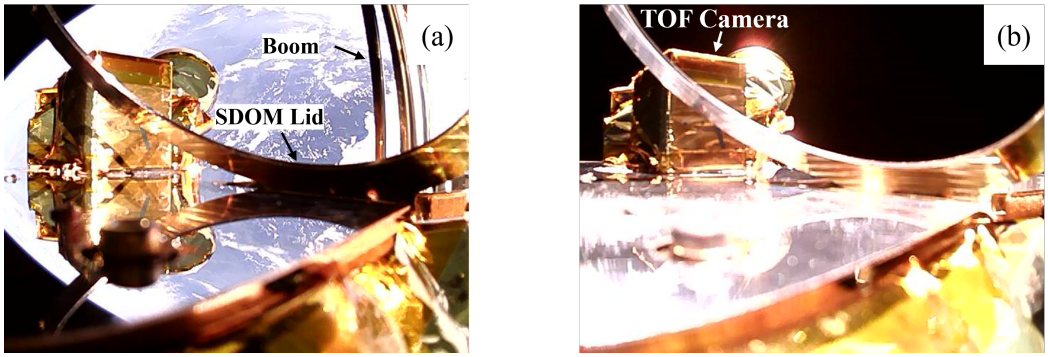


Figure 6. The separation of the boom and film of SDOM, as observed by DMC. (**a**) Before SDOM is separated. (**b**) After SDOM is separated.

3.2. Trouble of SDOM

The SDOM system broke two of the film's four corner connections in late 2019 and early 2020, effectively changing the film from a drag sail to a drag flag; the exact reason for the broken connections in these corners is still unknown. However, based on the observations, it is more likely that the Dyneema wires in the connections were untangled rather than severed. This situation could be observed by the DMC, as illustrated in Figure 7. The DMC was originally installed to monitor the conditions of the SDOM lid and boom extension.

ALE-1 is also equipped with a camera, called the Time-of-Flight (TOF) Camera System, for observing the SDOM film. This camera system collects distance information to the sail surface and downlinks the data to the ground to allow analyses to obtain better insight

into the three-dimensional dynamic behavior of the boom and drag sail in space [20]. In fact, we reported the results of our estimation of the shape of the SDOM when deployed in space in 2022 [21].

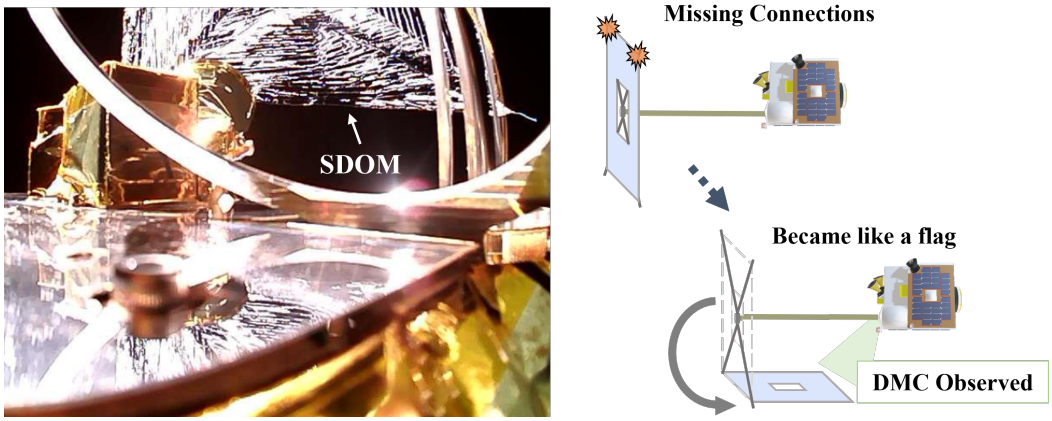


Figure 7. Disconnected SDOM film and its relative attitude as observed by DMC.

Now, the SDOM film with the two missing connections exhibited a characteristic behavior that was observed by the onboard cameras. It was determined that the film was exhibiting small movements around an axis through the remaining connections. Most of the time, the film remained on the opposite side of the DOM deployment plane in relation to the satellite, but there were instances where it was observed to move to the front side of the deployment plane. Figure 7 captured the film at the moment it unintentionally entered the field of view of the DMC while being positioned on the front side of the DOM deployment plane. The photograph clearly shows that one corner of the film is disconnected; the subsequent DMC photograph did not show the film at all. From this investigation, it was determined that the film can freely move around depending on the relative motion between the satellite's main structure.

Although the SDOM unintentionally became a drag flag, the orbital history shows that it was nevertheless effective for the orbital descent. This suggests that the SDOM, as it trailed like a cloak, experienced a certain level of atmospheric drag, which was expected to reduce the velocity of ALE-1. In this study, we examine the impact of this drag flag on the orbital descent by comparing the descent simulations and orbital history.

4. Investigation on Orbital Decent

4.1. Parameters of Orbit Analysis

We performed numerical simulations on the trajectory of ALE-1 with the SDOM film deployment. In general, the perturbed acceleration of a flying object in the upper atmosphere can be described by the following equation:

$$a_{\text{drag}} = -\frac{1}{2} C_D \rho \frac{A}{m} v_{\text{rel}}^2 \tag{1}$$

where ρ is the atmospheric density, C_D is the drag coefficient, A is the cross-sectional area of the satellite, m is the mass of the satellite, and v_{rel} is the relative velocity against the atmosphere. The mass of ALE-1 is 68.16 kg (65.29 kg after SDOM separation), as shown in Table 1, which is measured before launch. The C_D is a dimensionless quantity; for a satellite, it is commonly set to be 2.2 [22]. Although A is 6.25 m^2 at maximum when SDOM is deployed, the cross-sectional area value cannot be measured in the situation because the film is in a drag flag state. In addition, because C_D varies with the shape of the drag flag as well as the angles of attack, it is difficult to make a reliable assumption on the value of C_D. It is also difficult to calculate the C_D due to the unknown shape of the drag flag for the long period of the orbit transfer. Thus, C_D and A need to be handled as unknown parameters, which act as the limitation of the analysis.

According to this background, we decided to use the combination of these parameters $C_D \times A$ as an evaluation index. The range of $C_D A$ for the simulation analysis is determined based on the traditional fixed value of $C_D = 2.2$ and the mean cross-sectional area of ALE-1. The mean cross-sectional area of a tumbling satellite can be approximated as the sum of the projected areas in the six orthogonal directions (plus and minus directions in each of the three orthogonal axes) divided by four. As this value is about 3.4 m^2 for ALE-1 with the fully deployed SDOM and about 0.4 m^2 without SDOM, we set $C_D A_{ave}$ to be in the approximate range of 0.80–7.43.

For the initial orbit conditions, the TLE of ALE-1 on 28 May 2020 was used. The reason for not using the TLE immediately after the SDOM deployment is to avoid the influence of the gas leak, which caused an instantaneous increase in orbit altitude. Orbit propagation was performed by using the fourth-order RungeKutta method, with the solar and lunar mass perturbations, atmospheric drag, solar radiation pressure, and Earth gravity field being based on a non-spherical central body model [23]. The environmental models applied to the simulations are listed in Table 2, and the parameters used are summarized in Table 3. The atmospheric density model was calculated based on the parameters available on 1 February 2023. The simulator used for the analysis is the MEVIμS system, which was developed by our research team [24,25].

Table 2. Environmental models of the orbit analysis.

Environment	Model	References
Earth gravity model	EGM-08 (degree, order) = (40,40)	Ref. [22]
Sun and Moon model	DE431	Ref. [22,23]
Atmosphere model	NRLMSISE-00	Ref. [26]

Table 3. Parameters of the orbit analysis.

Items	Parameters	Value	References
Atmospheric drag	C_D	-	Ref. [22]
	A [m^2]	-	-
	$F_{10.7}$	Variable	Ref. [26]
	$F_{10.7\ 81\ days}$	Variable	Ref. [26]
Solar radiation	c_R [-] (reflectivity)	1.0	Ref. [22]
	P_{srp} [N/m^2] (solar-radiation pressure)	4.54 E-6	Ref. [22]
Time system	ΔAT [s] (delta atomic time)	37.0	Ref. [27]
Satellite properties	Mass of satellite [kg] (after SDOM separation)	65.29	-
	C_DA [m^2]	0.80 to 7.43	-

4.2. Evaluation of SDOM De-Orbiting Capability

The results of the overall analysis are shown in Figure 8 together with the orbit altitude information of ALE-1 based on the TLE. In 2020, the orbit of ALE-1 with $C_D A = 3.0$ m^2 was in good agreement with the TLE. However, in 2021, the descent began to accelerate, causing the orbit to significantly change from $C_D A = 1.5$ m^2 to $A = 4.0$ m^2 over the course of the year until the SDOM was finally separated. It is worth noting that the $C_D A$ of the drag flag increased as the orbital altitude decreased. This result is a very important finding on the de-orbiting performance of a drag flag attached to a satellite in low Earth orbit (LEO). The value of $C_D A = 4.0$ m^2 is corresponding to a mean cross-sectional area of $A = 1.8$ m^2 if the drag coefficient C_D is approximated as 2.2.

As discussed in Section 3.2, the SDOM was observed to be freely moving around a single axis, fluttering like a flag. Even in this state, the drag flag delivered a de-orbiting effect on the order of $C_D A = 3.0$ m^2 to 4.0 m^2, or approximately 22 to 29% of the possible maximum value of the scenario where the drag sail is always set vertical to the velocity

vector, and approximately 41 to 54% of the mean value of the scenario where the satellite is freely tumbling.

This result indicates that the shape of the de-orbiting film does not necessarily need to be fixed as a flat surface, as was regarded as the requirement so far worldwide; instead, a trailing drag flag can be about up to half as effective as a flat drag sail in LEO in the case of ALE-1 implementation. This new finding can open up breakthrough opportunities for more effective ways of implementing PMD (Post-Mission Disposal) devices in terms of their simplicity, size, mass, and reliability.

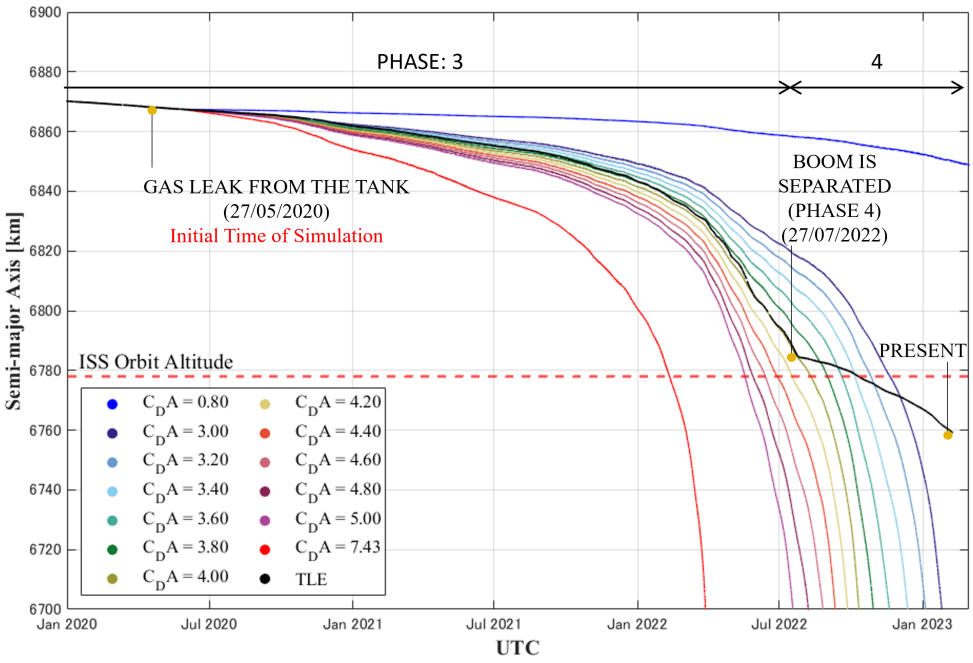


Figure 8. Comparison between the flight data and orbit propagation simulations.

The trailing drag flag is expected to move backwards with the increase in atmospheric density and is expected to stabilize in a cloak-like manner, resulting in a smaller effective cross-sectional area; in other words, ρ and A may be inversely related. The above-mentioned increasing effective cross-sectional area along the descent of the orbit can suggest that the complex behavior of the drag flag in the rarefied atmosphere may have resulted in a three-dimensional shape, which in turn resulted in the increasing mean atmospheric perturbation. It was also reported in the past study that even a flat surface that is parallel to the velocity vector delivers a non-zero atmospheric drag coefficient due to the thermochemical properties of the upper atmosphere [28]. It can be that this effect also helped the drag flag to achieve the above-mentioned de-orbiting performance. Details on this topic needs to be investigated further in the future.

5. Conclusions

ALE-1, the technology demonstration satellite for artificial shooting star generation, successfully conducted an orbital descent maneuver from an initial orbit of about 500 km down to about 400 km with the help of an SDOM system. ALE-1 also succeeded at remaining in the target orbit for a longer period to conduct its mission by separating the DOM. In addition, ALE-1 also succeeded at obtaining camera images of several different configurations of the SDOM during the orbital operation.

The SDOM, however, experienced mechanical troubles in space, and the drag sail resulted in a drag flag; nevertheless, the orbit of the ALE-1 was able to be lowered as planned, although it took longer than expected, indicating that the drag flag shape has an effective de-orbiting performance. Consequently, a comprehensive investigation was

conducted on the de-orbiting capabilities of the drag flag by comparing orbital propagation simulation results and the flight data of the satellite altitude over the past three years of operation. A combination of parameters $C_D A$ was used to evaluate the de-orbiting performance of the SDOM, which can be regarded as the mean values of the drag coefficient multiplied by the cross-sectional area. In this way, the mean de-orbiting performance could be evaluated without an explicit analysis on the drag coefficient, which can be affected by the shape of the fluttering drag flag in the rarefied atmosphere in the orbit. Through the investigation, the following points can be found:

- The $C_D A$ of the drag flag increased as the orbital altitude decreased;
- The $C_D A$ of the drag flag was estimated to be ranging from 3.0 to 4.0, which corresponds to about 22 to 29% of the $C_D A$ of the fully deployed DOM and 39 to 51% of the $C_D A$ of a tumbling satellite with the deployed DOM;
- Drag flags can be effective de-orbit devices and can provide breakthroughs for future PMD devices to solve space debris problems.

The exact behaviors of the drag flag and drag sail attached to satellites are subject to further investigations; the above findings, however, indicate that materials in different shapes than sails, such as threads, tapes, and mantles, may have similar de-orbiting performances and can be utilized for future PMD devices to solve space debris problems. In addition, de-orbiting devices based on these various shapes can have possibilities to be implemented more mechanically efficiently, resulting in more lightweight and small solutions. It is therefore also possible that significantly larger and higher performance de-orbiting devices can be developed with reduced mass and envelope.

From the satellite system design point of view, the developed de-orbiting device SDOM system has several superior benefits compared with the traditional methods of de-orbiting using thruster systems. Firstly, SDOM's only 4 kg of mass is favorable for micro-satellites with a mass of approximately 100 kg or less. Secondly, it does not contain hazardous materials such as propellant and can be handled very safely on the ground, during the launch, and even possibly inside manned spacecraft. Finally, it does not necessitate active attitude control or power consumption during the de-orbiting as demonstrated in this study; hence, it can ensure that the satellite can de-orbit, even if the satellite is no longer operational after the activation of the device, which is of great benefit as a de-orbiting device. The authors sincerely hope that this research's results can contribute to the enhancement of future peaceful space utilization.

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